```
All libraries imported successfully!
```

```
CSV dataset loaded.
Dataset shape: (5572, 5)
```

	v1	v2	Unnamed: 2	Unnamed: 3	Unnamed: 4
0	ham	Go until jurong point, crazy.. Available only ...	NaN	NaN	NaN
1	ham	Ok lar... Joking wif u oni...	NaN	NaN	NaN
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	NaN	NaN	NaN
3	ham	U dun say so early hor... U c already then say...	NaN	NaN	NaN
4	ham	Nah I don't think he goes to usf, he lives aro...	NaN	NaN	NaN

```
Original shape: (5572, 5)

Missing values before cleaning:
v1              0
v2              0
Unnamed: 2    5522
Unnamed: 3    5560
Unnamed: 4    5566
dtype: int64

Shape after cleaning: (5572, 2)

Column names: ['label', 'message']
```

	label	message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

```
Dataset shape: (5572, 2)

Missing values:
label      0
message    0
dtype: int64

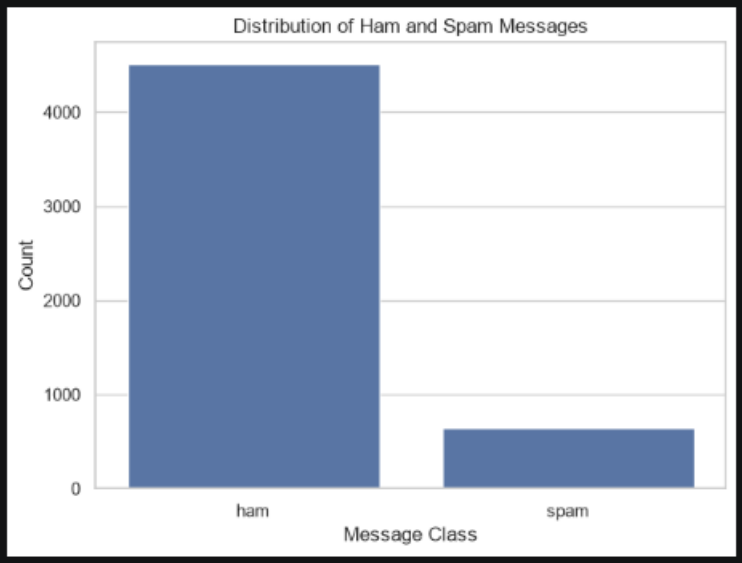
Label counts:
label
ham     4825
spam     747
Name: count, dtype: int64

Unique labels: <ArrowStringArray>
['ham', 'spam']
Length: 2, dtype: str
Duplicate rows: 403
```

```
Shape after cleaning: (5158, 2)
Missing values:
 label      0
message    0
dtype: int64
Labels:
 label
ham     4516
spam     642
Name: count, dtype: int64
```

```
Class counts:
 label
ham     4516
spam     642
Name: count, dtype: int64

Class percentages:
 label
ham     87.55
spam    12.45
Name: proportion, dtype: float64
```



	message	clean_message	label
0	Go until jurong point, crazy.. Available only ...	go until jurong point crazy available only in ...	ham
1	Ok lar... Joking wif u oni...	ok lar joking wif u oni	ham
2	Free entry in 2 a wkly comp to win FA Cup fina...	free entry in a wkly comp to win fa cup final ...	spam
3	U dun say so early hor... U c already then say...	u dun say so early hor u c already then say	ham
4	Nah I don't think he goes to usf, he lives aro...	nah i dont think he goes to usf he lives aroun...	ham
5	FreeMsg Hey there darling it's been 3 week's n...	freemsg hey there darling its been weeks now a...	spam
6	Even my brother is not like to speak with me. ...	even my brother is not like to speak with me t...	ham
7	As per your request 'Melle Melle (Oru Minnamin...	as per your request melle melle oru minnaminun...	ham
8	WINNER!! As a valued network customer you have...	winner as a valued network customer you have b...	spam
9	Had your mobile 11 months or more? U R entitle...	had your mobile months or more u r entitled to...	spam

```
Training samples: 4126
Testing samples: 1032

Training label distribution:
label
ham     3612
spam     514
Name: count, dtype: int64
```

```
Naive Bayes training completed.
Predictions: ['ham' 'ham' 'ham' 'ham' 'ham' 'ham' 'ham' 'ham' 'ham' 'ham']
```

```
Naive Bayes Results
Accuracy : 0.9690
Spam Precision: 1.0000
Spam Recall   : 0.7500
Spam F1-score : 0.8571

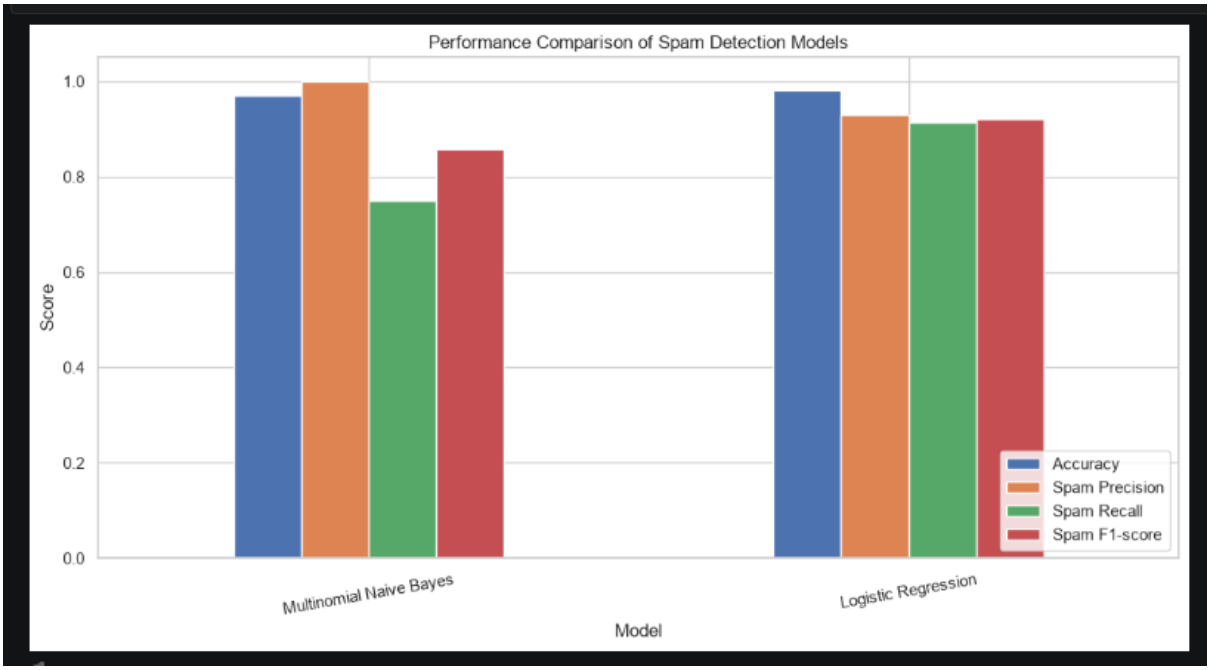
Classification Report:
              precision    recall  f1-score   support

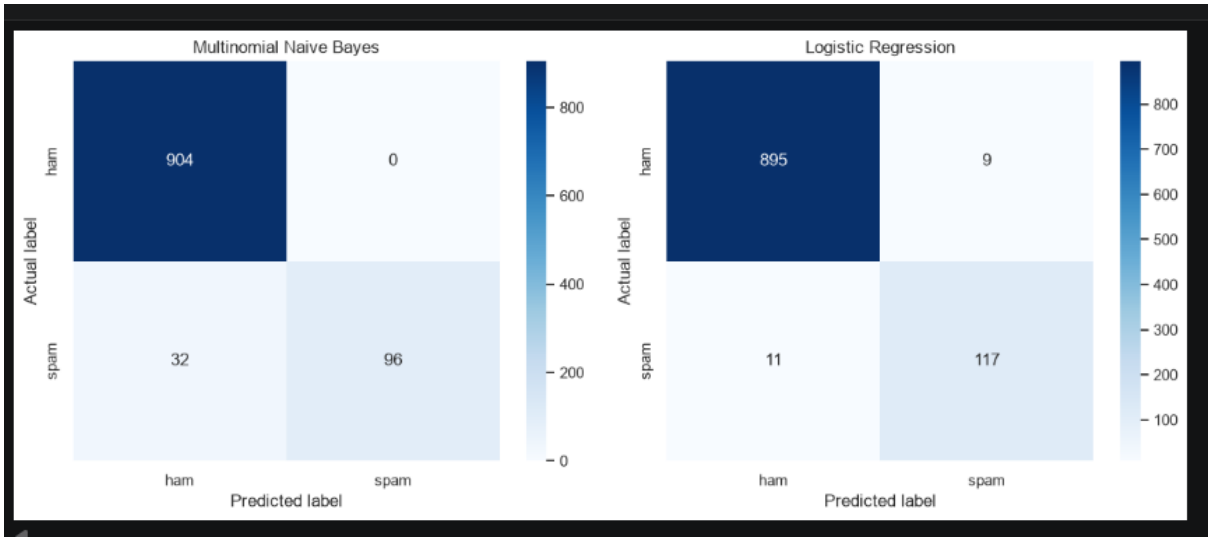
         ham       0.97      1.00      0.98       904
        spam       1.00      0.75      0.86       128

    accuracy                           0.97      1032
   macro avg       0.98      0.88      0.92      1032
weighted avg       0.97      0.97      0.97      1032
```

```
Logistic Regression training completed.
```

	Model	Accuracy	Spam Precision	Spam Recall	Spam F1-score
0	Multinomial Naive Bayes	0.9690	1.0000	0.7500	0.8571
1	Logistic Regression	0.9806	0.9286	0.9141	0.9213






```
Selected model: Logistic Regression
Total misclassified messages: 20
```

	Actual	Predicted	Message
18	ham	spam	Ok... Ur typical reply...
77	spam	ham	Cashbin.co.uk (Get lots of cash this weekend!)...
146	ham	spam	Call me da, i am waiting for your call.
148	ham	spam	Madam,regret disturbance.might receive a refer...
164	ham	spam	Plz note: if anyone calling from a mobile Co. ...
205	spam	ham	Sorry I missed your call let's talk when you h...
241	ham	spam	Host-based IDPS for linux systems.
323	spam	ham	Sorry! U can not unsubscribe yet. THE MOB offe...
354	spam	ham	Hi I'm sue. I am 20 years old and work as a la...
389	spam	ham	Babe: U want me dont u baby! Im nasty and have...
625	ham	spam	Pete,is this your phone still? Its Jenny from ...
657	ham	spam	Hi! You just spoke to MANEESHA V. We'd like to...
835	spam	ham	Monthly password for wap. mobsi.com is 391784....
848	spam	ham	dating:i have had two of these. Only started a...
863	spam	ham	Guess who am I?This is the first time I create...

```
Message: Congratulations! You have won a free prize. Claim now!
Prediction: SPAM
------------------------------------------------------------
Message: Hey, are we meeting for lunch today?
Prediction: HAM
------------------------------------------------------------
Message: URGENT! You have won cash. Call now to claim your reward.
Prediction: SPAM
------------------------------------------------------------
Message: I will reach home in ten minutes.
Prediction: HAM
------------------------------------------------------------
```

```
Best model based on spam F1-score:
Model              Logistic Regression
Accuracy                       0.98062
Spam Precision                0.928571
Spam Recall                   0.914062
Spam F1-score                  0.92126
Name: 1, dtype: object

Selected model: Logistic Regression
```

```
Model saved as: sms_spam_detection_model.pkl
```

```
========== FINAL PROJECT CHECK ==========
Dataset shape: (5158, 3)
Dataset columns: ['label', 'message', 'clean_message']
Missing values: 0
Duplicate rows remaining: 0
Training samples: 4126
Testing samples: 1032
Models trained: Multinomial Naive Bayes, Logistic Regression
Best model: Logistic Regression
Best model spam F1-score: 0.9213
Saved model exists: True
=========================================
```